

BBA 2421

Business Mathematics

Lecture 3: Principles of Interest



Simple interest

- **Interest** is the price paid for borrowing money.
- Interest rates are usually expressed as a percent of the amount borrowed.
- There are basically two types of interest that are in common use today:
 - i. **Simple interest** and
 - ii. **Compound interest**.
- The interest that is paid only on the principal is called the **simple interest**.
- **Compound interest** on the other hand is interest that is paid on both the ***principal*** and ***past interest***.



Key terms

- **Principal**
 - the amount borrowed.
- **Rate (or rate of interest)**
 - percent of interest charged by lender.
- **Time**
 - the length of time in years before repayment is due.



Calculating simple interests

- Simple interest is interest charged on the entire principal for the entire length of the loan and is usually used for short-term loans that last less than a year.
- Simple interest is calculated using the formula below:

$$\textit{Simple interest} = \textit{Principal} \times \textit{Rate} \times \textit{Time}$$

$$I = P \times R \times T \text{ or}$$

$$I = PRT$$

- The rate, R , is expressed as a decimal or fraction, and time, T , is expressed as the number of years, or the fraction of a year.
- **Note:** Time (T) is measured in years. *Hence a time period given in months or days must be converted to a fraction of a year before being substituted into the formula for T .*



Maturity value

- When the loan matures you must pay the principal together with the interest.
- The amount that must be repaid when the loan is due is called the **maturity value**.
- The maturity value is also sometimes called the **future value** while the principal amount received by the borrower when the loan is made is called the **present value**.
- The maturity value of a loan always exceeds the principal (the original loan amount) since maturity value is principal plus interest.



- The maturity value is found using the following:

$$\text{Maturity value} = \text{Principal} + \text{Interest}$$

$$M = P + I$$

$$M = P + PRT, \text{ since } I = PRT$$

$$M = P(1 + RT)$$

In-class exercise

Suppose an individual borrows an amount of K28,300 for 10 months at an interest rate of 9.25%. Find:

- a) The interest due on the loan
- b) The maturity value at the end of 10 months.



Solution

a) Interest (I) = PRT

$$P = 28300, R = \frac{9.25}{100} = 0.0925, \text{ and } T = \frac{10}{12} = \frac{5}{6}$$

$$I = 28300 \times 0.0925 \times \frac{5}{6} = K2181.46$$

b) $M = P + I = K28300 + K2181.46 = K30481.46.$



Finding exact and ordinary interest

- Since the simple interest rate (r) is given per year, time must also be given in years or expressed as a fraction of a year when using $I=PRT$
- Therefore, if time is given in days, first convert it to a fraction of a year using:

$$T = \frac{\text{number of days in loan period}}{\text{number of days in a year}}$$

- Exact interest computations require the use of the exact number of days in a year, 365 or 366 if it's a leap year.



- For ordinary or bankers' interest, 360 days is used. This is because banks used 360 days in a year for interest rate calculations before calculators and computers became widely available.
- Note that some banks and financial institutions still use 360 days.



Calculating principal, rate and time

- We know that, $I = PRT$
- Therefore, $principal(P) = \frac{Interest(I)}{Rate(R) \times Time(T)}$

$$P = \frac{I}{RT}$$

In-class exercise

Suppose an individual borrows money at 10% for 54 days. Find the principal amount that results in interest of k780.



Solution

$$R = 0.1, T = \frac{54}{365} \text{ and } I = K780$$

$$P = \frac{I}{RT} = \frac{780}{0.1 \times \left(\frac{54}{365}\right)} = K52722.22$$

- To find rate use the following:

$$R = \frac{I}{PT}$$



In-class exercise

Suppose an individual leaves extra cash amounting to K86500 in an account from June 1 to August 16. Find the rate if the individual earned K365.22 in interest during this period of time.

Solution

Use the knuckle method to find the exact number of days.

30 days in June

-1 day of deposit in June

=29 days left in June

+ 31 days in July

=60 days



+ 16 first 16 days in August

=76 days from June 1 to August 16

- Therefore, Time (T) = $\frac{76}{360}$

$$R = \frac{365.22}{86500 \times \frac{76}{360}} = 0.02.$$

- To find the time, use

$$\text{Time} = \frac{\text{Interest}}{\text{Principal} \times \text{Rate}}$$

$$T = \frac{I}{PI}$$



- Note that the preceding formula gives the time in years.
- However, we often need time in days or months and thus we use the following alternative formulas:

$$\textit{Time in days} = \frac{I}{PR} \times 365 \textit{ for exact interest}$$

$$\textit{Time in days} = \frac{I}{PR} \times 360 \textit{ for ordinary or banker's interest}$$

$$\textit{Time in months} = \frac{I}{PR} \times 12$$



In-class exercise

Suppose an individual deposited k6,200 in an account paying 3% and earned k72.33 in interest. Find the number days that the deposit earned interest (Round to a whole number of days)

Solution

$$\begin{aligned} T \text{ in days} &= \frac{I}{PR} \times 360 \\ &= \frac{72.33}{6200 \times 0.03} \times 360 \\ &= 140 \text{ days (rounded)} \end{aligned}$$

Therefore, the money was deposited for 140 days.



Compound Interest

- **Simple interest** is paid on the principal—not on any past interest.
- **Compound interest** is calculated on any interest previously credited (paid) to the account in addition to the original principal.
 - Compound interest earns interest on principal and it also earns interest on any interest already earned.
- Let's assume an account pays interest at a rate of r per year.
- Now if interest is compounded yearly, then after year 1, the earned interest is added to the original principal amount to define a larger principal base in year 2.



- Therefore, during year 2, the account earns interest on interest.
- We note that the money left in the account grows by $(1 + r)$ after 1 year, and after two years the amount grows by another factor of $(1 + r)$ to $(1 + r)^2$.
- Thus, after n years, the account will grow to $(1 + r)^n$ times its original value.
- In other words, each time the compound interest is calculated, it is added to the principal amount prior to calculating interest again.
- Thus, the compound interest is calculated on the principal plus any interest previously credited/added to the account.
- In practice, compound interest is applied to savings accounts, money market accounts, certificates of deposits, and retirement accounts



- Let us illustrate this concept in the table below:

Year	Starting Principal	Interest earned	Ending principal
1	P	rP	$P + rP = P(1 + r)$
2	$P(1 + r)$	$rP(1 + r)$	$P(1 + r) + rP(1 + r)$ $= (1 + r)(P + rP)$ $= P(1 + r)^2$
3	$P(1 + r)^2$	$rP(1 + r)^2$	$P(1 + r)^2 + rP(1 + r)^2$ $= P(1 + r)^3$
-	-	-	-
n	$P(1 + r)^{n-1}$	$rP(1 + r)^{n-1}$	$P(1 + r)^n$



COMPOUNDING PERIOD AND NUMBER OF COMPOUND PERIODS

- **Compounding period** is the time over which interest is calculated and added to principal.
 - For example, $K\%$ compounded monthly means that the interest will be calculated and added to the principal at the end of each month.
- **Number of compounding** periods is the number of compounding periods, either in each year or in the life of a loan or investment.
- The table below sheds more light:



Interest compounded	Compound at the end of every	Number of compounding periods in one year
Semi-annually	6 months	2
Quarterly	3 months	4
Monthly	1 month	12
Daily	1 day	365

- Note that leap year has 366 compounding periods.



- **Interest rate per compounding period (i)** is the interest rate applied to each compounding period.
 - It is found by dividing the annual interest rate by the number of compounding periods in a year.
 - The total number of compounding periods is the number per year times the number of years.
- Interest rate per compounding period is given by

$$i = \frac{\text{Interest rate per year}}{\text{Number of compounding periods in a year}}$$



Rate	compounded	Number of compounding periods per year	Term	Rate per compounding period	Total number of compounding periods
8%	semiannually	2	3 years	$\frac{8\%}{2}=4\%$	3 years×2 per year=6
12%	Monthly	12	2.5 years	$\frac{12\%}{12}=1\%$	2.5years×12 per year=30
9%	Quarterly	4	5 years	$\frac{9\%}{4}=2.25\%$	5 years×4 per year=20

- The compounded amount or maturity value is given by:

$$M = P[1 + i]^n$$

Where i =interest rate per compounding period, P = initial investment,
 n = total number of compounding periods (i.e. time×number of compounding periods) and M =maturity value.



- The interest earned will be given by $I = M - P$

In-class exercise

An investment managed by Bank of America pays 7% interest per year compounded semi-annually. Given an initial deposit of $K4500$, Find:

- (a) The compound amount after 5 years, and
- (b) find the compound interest.



Solution

(a) Interest is compounded at $\frac{7\%}{2} = 3.5\%$ every 6 months

Total number of compounding periods = 5 years $\times$ 2 periods per year
= 10 periods

Therefore, 3.5% is the interest rate per compounding period (i) and 10 is the number of compounding periods (n).

$$\begin{aligned} M &= P(1 + i)^n \\ &= K4500 \times (1 + 0.035)^{10} \\ &= K4500 \times (1.035)^{10} \\ &= K6347.69 \text{ (rounded)} \end{aligned}$$

The compound amount is K6347.69.



$$(b) I = M - P = K6347.69 - K4500 = K1847.69$$



Bank Discounts

- **Simple interest notes** involve principal (face value or loan amount), interest rate, time, interest, and maturity value.
- Simple discount notes involve **proceeds (loan amount)**, **discount rate**, time, **bank discount (interest)**, and **face value (or maturity value)**.
- Face value in a simple interest note is the amount loaned to the borrower, but it is the maturity value in a simple discount note.
- Simple discount notes are also called **interest-in-advance notes**, since interest is subtracted before funds are given to the borrower.
- A basic difference between the two types of notes is that simple interest is calculated based on principal, whereas simple discount is calculated based on maturity value.



Simple Interest versus Simple Discount Notes

Type of Note	Loan Amount		Interest		Repayment Amount
Simple interest	Face value (Principal)	+	Interest	=	Maturity value
Simple discount	Proceed	+	Discount (Interest)	=	Face value (Maturity value)

- A bank discount can be defined as the interest charge on the loan or the interest that must be paid at maturity.
- The bank discount/interest, which is based on simple discount information, is calculated using the following formula:

$$\text{Bank discount} = \text{Face value} \times \text{Discount rate} \times \text{Time}$$



- That is

$$B = M \times D \times T$$

- Then, if P is the proceeds to the borrower,

Proceeds (loan amount) = Face value – Bank discount or

$$P = M - B$$

- Stated in another way,

Face value = Proceeds (loan amount) + Bank discount or

$$M = P + B$$

- *Note: Time must be given in years*



In-class exercise

Bambo Nyasulu signs a simple discount note with a face or maturity value of K35,000 so that he can purchase a truck. The banker discounts the 10-month note at 9%. Find the amount of the discount and the proceeds.



Solution

Bambo Nyasulu does not receive K35,000 from the bank—that is the amount he must repay when the loan matures. Use $M = K35,000$, $D = 9\%$, and $T = 1 \frac{10}{12}$ in the formula $B = MDT$ to find the discount, which is the interest that must be paid at maturity.

$$\text{Bank discount} = M \times D \times T = K35000 \times 0.09 \times \frac{10}{12} = K2625$$

- The discount of K2625 is the interest charge on the loan. The proceeds that Peterson actually receives when making the loan is found using $P = M - B$.

$$P = M - B = K35,000 - K2625 = K32,375$$

- Bambo Nyasulu receives K32,375 but must repay K35,000 when the loan matures in 10 months. The difference is the interest.



Finding the face value

- Calculating Face Value to Achieve Desired Proceeds

$$M = \frac{P}{1 - DT}$$

- Where

M = Face value of the simple discount note

P = Proceeds received by the borrower

D = Discount rate used by the bank

T = Time of the loan (in years)

Note:

- The symbol D is the discount rate, not the bank discount.



Example

- Tina Watson plans to rebuild her house. She estimates that she will need to borrow k18,000 for 180 days. Find the face value of the 10% simple discount note that would result in proceeds of K18,000 to Watson.

Solution

$$M = \frac{P}{1 - DT}$$

$$P = 18000, D = 0.1 \quad T = \frac{180}{360}$$

- Therefore, $M = \frac{18,000}{1 - \left(0.10 \times \frac{180}{360}\right)} = K18,947.37$



In-class exercise

A simple 300-day simple discount note has proceeds of K48,000 and a discount rate of 8.8%. Find the maturity value.



End of Lecture 3

